

Comparative of Extended and Unscented Kalman Filters for Radar-Based 2D Position Estimation

*Note: This work builds upon Alex Becker's 'Kalman Filter from the Ground Up' [1].

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Abstract—Nonlinear state estimation is fundamental to numerous applications in robotics, aerospace, and autonomous systems. While the Extended Kalman Filter (EKF) has been the de facto standard for decades, its linearization approach can introduce significant errors in highly nonlinear systems. This paper presents a comprehensive comparative analysis of the EKF and Unscented Kalman Filter (UKF) in the context of radar-based vehicle tracking under non-Gaussian measurement noise. We demonstrate that the UKF's sigma point approach provides superior covariance propagation through nonlinear transformations, particularly in polar-to-Cartesian coordinate conversions. Through Monte Carlo simulations and realistic radar tracking scenarios with complex maneuvers, we show that UKF achieves substantial improvements in position estimation accuracy over EKF. Our results indicate that despite moderate computational overhead, the UKF's robustness to linearization errors and ability to handle non-Gaussian noise make it the preferred choice for safety-critical tracking applications where accuracy and reliability are paramount.

Index Terms—Extended Kalman Filter, Unscented Kalman Filter, Nonlinear Estimation, Radar Tracking, State Estimation, Sigma Points, Covariance Propagation

I. INTRODUCTION

State estimation in nonlinear systems remains one of the most challenging problems in control theory and signal processing. The Extended Kalman Filter (EKF), introduced by Stanley Schmidt for the Apollo program [2], has been the workhorse of nonlinear estimation for over five decades. However, its reliance on first-order Taylor series linearization can lead to significant estimation errors and potential filter divergence in highly nonlinear systems.

The Unscented Kalman Filter (UKF), proposed by Julier and Uhlmann [3], addresses these limitations through a deterministic sampling approach called the unscented transform. Rather than linearizing the nonlinear functions, the UKF propagates carefully chosen sigma points through the true nonlinear transformation, providing a more accurate representation of the mean and covariance.

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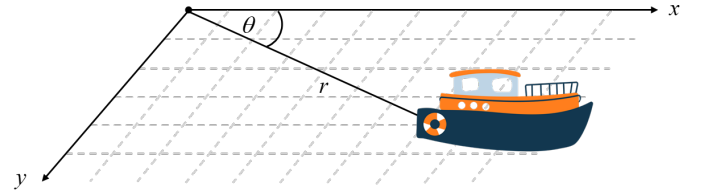


Fig. 1. Radar measurement geometry. The sensor measures range r and bearing angle θ to the target vehicle in polar coordinates. The nonlinear transformation to Cartesian coordinates (x, y) introduces significant challenges for linearization-based filtering approaches, motivating the comparison between EKF and UKF methodologies.

A. Problem Statement

This work addresses the specific challenge of radar-based vehicle tracking, where measurements arrive in polar coordinates (range and bearing) but state estimation is performed in Cartesian space. As illustrated in Figure 1, the radar sensor measures range r and bearing angle θ to the target vehicle, which must be transformed to Cartesian coordinates (x, y) for state estimation. This polar-to-Cartesian transformation is inherently nonlinear, making it an ideal test case for comparing linearization-based and sigma point-based approaches. Additionally, we introduce realistic non-Gaussian measurement noise to evaluate filter robustness beyond ideal conditions.

II. THEORETICAL BACKGROUND

A. The State Estimation Problem

State estimation addresses the fundamental challenge of determining the internal states of a dynamic system when only indirect, noisy measurements are available. The problem is formulated as a recursive Bayesian inference task where we seek to estimate the posterior probability distribution of the state given all available measurements.

Consider the discrete-time nonlinear system described by the state transition equation:

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}) + \mathbf{w}_k \quad (1)$$

and the measurement equation:

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k \quad (2)$$

Here, $\mathbf{x}_k \in \mathbb{R}^n$ represents the state vector at time step k . The function $f(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ describes the system dynamics, while the measurement vector $\mathbf{z}_k \in \mathbb{R}^m$ is related to the true state through the observation function $h(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The process noise $\mathbf{w}_k$ and measurement noise $\mathbf{v}_k$ are assumed to be zero-mean Gaussian random variables with covariances $\mathbf{Q}_k$ and $\mathbf{R}_k$ respectively:

$$\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q}_k), \quad \mathbf{v}_k \sim \mathcal{N}(0, \mathbf{R}_k) \quad (3)$$

The Kalman filter provides an optimal recursive solution when both $f(\cdot)$ and $h(\cdot)$ are linear functions. However, when these functions are nonlinear, approximation techniques are required. The central difficulty arises from the propagation of probability distributions through nonlinear functions. When a Gaussian-distributed random variable passes through a nonlinear transformation, the resulting distribution is generally non-Gaussian, violating the Gaussian assumption fundamental to the Kalman filter framework.

B. Extended Kalman Filter: Analytical Linearization

1) *Taylor Series Approximation:* The Extended Kalman Filter addresses nonlinearity through first-order Taylor series linearization. For a nonlinear function $\mathbf{y} = g(\mathbf{x})$, the Taylor series expansion around point $\mathbf{a}$ yields:

$$g(\mathbf{x}) \approx g(\mathbf{a}) + \left. \frac{\partial g}{\partial \mathbf{x}} \right|_{\mathbf{a}} (\mathbf{x} - \mathbf{a}) \quad (4)$$

where $\left. \frac{\partial g}{\partial \mathbf{x}} \right|_{\mathbf{a}}$ is the Jacobian matrix, retaining only first-order terms.

2) *Jacobian Matrices:* The state transition Jacobian is:

$$\mathbf{F}_k = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k-1|k-1}} \quad (5)$$

The measurement Jacobian is:

$$\mathbf{H}_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k|k-1}} \quad (6)$$

3) *EKF Algorithm:* The prediction step propagates the state and covariance:

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}) \quad (7)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}_k \mathbf{P}_{k-1|k-1} \mathbf{F}_k^T + \mathbf{Q}_k \quad (8)$$

The update step incorporates measurements through:

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k \quad (9)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad (10)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})) \quad (11)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (12)$$

where $\mathbf{S}_k$ is innovation covariance and $\mathbf{K}_k$ is Kalman gain.

C. Unscented Kalman Filter: Statistical Linearization

1) *The Unscented Transform:* The Unscented Kalman Filter approximates the probability distribution rather than the nonlinear functions. For a random variable $\mathbf{x}$ with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{P}_x$ transformed through $\mathbf{y} = g(\mathbf{x})$, the unscented transform: (1) selects $2n + 1$ sigma points that capture the statistics of $\mathbf{x}$, (2) propagates each point through $g(\cdot)$, and (3) reconstructs the statistics of $\mathbf{y}$ using weighted averages.

2) *Sigma Point Selection:* For an n -dimensional state with mean $\bar{\mathbf{x}}$ and covariance $\mathbf{P}$:

$$\chi^{(0)} = \bar{\mathbf{x}} \quad (13)$$

$$\chi^{(i)} = \bar{\mathbf{x}} + \left(\sqrt{(n + \lambda) \mathbf{P}} \right)_i, \quad i = 1, \dots, n \quad (14)$$

$$\chi^{(i)} = \bar{\mathbf{x}} - \left(\sqrt{(n + \lambda) \mathbf{P}} \right)_{i-n}, \quad i = n + 1, \dots, 2n \quad (15)$$

where $\left(\sqrt{(n + \lambda) \mathbf{P}} \right)_i$ is the i -th column of the matrix square root (Cholesky decomposition), and:

$$\lambda = \alpha^2(n + \kappa) - n \quad (16)$$

Typical values: $\alpha \in [10^{-4}, 1]$ and $\kappa = 0$ or $\kappa = 3 - n$.

3) *Weight Assignment:* Mean weights:

$$W_0^{(m)} = \frac{\lambda}{n + \lambda}, \quad W_i^{(m)} = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n \quad (17)$$

Covariance weights (with $\beta = 2$ for Gaussian distributions):

$$W_0^{(c)} = \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta), \quad W_i^{(c)} = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n \quad (18)$$

4) *UKF Prediction Step:* Generate sigma points from the previous posterior:

$$\chi_{k-1}^{(i)} = \text{SigmaPoints}(\hat{\mathbf{x}}_{k-1|k-1}, \mathbf{P}_{k-1|k-1}), \quad i = 0, \dots, 2n \quad (19)$$

Propagate through the state transition function:

$$\chi_{k|k-1}^{(i)} = f(\chi_{k-1}^{(i)}), \quad i = 0, \dots, 2n \quad (20)$$

Compute predicted state and covariance:

$$\hat{\mathbf{x}}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(m)} \chi_{k|k-1}^{(i)} \quad (21)$$

$$\mathbf{P}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(c)} [\chi_{k|k-1}^{(i)} - \hat{\mathbf{x}}_{k|k-1}] [\chi_{k|k-1}^{(i)} - \hat{\mathbf{x}}_{k|k-1}]^T + \mathbf{Q}_k \quad (22)$$

5) *UKF Update Step:* Transform sigma points to measurement space:

$$\mathcal{Z}_{k|k-1}^{(i)} = h(\chi_{k|k-1}^{(i)}), \quad i = 0, \dots, 2n \quad (23)$$

Compute measurement statistics:

$$\hat{\mathbf{z}}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(m)} \mathcal{Z}_{k|k-1}^{(i)} \quad (24)$$

$$\mathbf{P}_{zz} = \sum_{i=0}^{2n} W_i^{(c)} [\mathcal{Z}_{k|k-1}^{(i)} - \hat{\mathbf{z}}_{k|k-1}] [\mathcal{Z}_{k|k-1}^{(i)} - \hat{\mathbf{z}}_{k|k-1}]^T + \mathbf{R}_k \quad (25)$$

$$\mathbf{P}_{xz} = \sum_{i=0}^{2n} W_i^{(c)} [\chi_{k|k-1}^{(i)} - \hat{\mathbf{x}}_{k|k-1}] [\mathcal{Z}_{k|k-1}^{(i)} - \hat{\mathbf{z}}_{k|k-1}]^T \quad (26)$$

Update state and covariance:

$$\mathbf{K}_k = \mathbf{P}_{xz} \mathbf{P}_{zz}^{-1} \quad (27)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \hat{\mathbf{z}}_{k|k-1}) \quad (28)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{P}_{zz} \mathbf{K}_k^T \quad (29)$$

D. Comparative Analysis

The fundamental distinction between EKF and UKF lies in their treatment of nonlinearity. The EKF linearizes the functions using first-order Taylor expansion, whereas the UKF approximates the probability distribution using deterministic sampling. The EKF achieves first-order accuracy in the mean and second-order accuracy in the covariance, while the UKF achieves second-order accuracy in the mean and third-order accuracy in the covariance for Gaussian distributions.

Computationally, the EKF requires $O(n^3)$ operations per iteration, dominated by covariance updates and Jacobian computations. The UKF requires $O(n^3 + (2n+1)T_f)$ operations, where T_f is the cost of evaluating $f(\mathbf{x})$ or $h(\mathbf{x})$. Table I provides a quantitative comparison of key algorithmic properties.

TABLE I
EKF VS UKF ALGORITHMIC COMPARISON

Property	EKF	UKF
Approximation method	Taylor series	Unscented transform
Mean accuracy	$O(\Delta t^2)$	$O(\Delta t^3)$
Covariance accuracy	First-order	Second-order
Jacobian required	Yes	No
Function evaluations	1 per step	$2n+1$ per step
Derivative computation	Analytical	Numerical

III. EXPERIMENTAL METHODOLOGY

A. System Dynamics

The state vector contains position, velocity, and acceleration in 2D:

$$\mathbf{x} = [x, \dot{x}, \ddot{x}, y, \dot{y}, \ddot{y}]^T \quad (30)$$

The state transition model follows constant acceleration dynamics:

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} & 0 & 0 & 0 \\ 0 & 1 & \Delta t & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & \Delta t & \frac{\Delta t^2}{2} \\ 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (31)$$

Process noise covariance models acceleration uncertainty:

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & \frac{\Delta t^2}{2} & 0 & 0 & 0 \\ \frac{\Delta t^3}{2} & \Delta t^2 & \Delta t & 0 & 0 & 0 \\ \frac{\Delta t^2}{2} & \Delta t & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\Delta t^4}{4} & \frac{\Delta t^3}{2} & \frac{\Delta t^2}{2} \\ 0 & 0 & 0 & \frac{\Delta t^3}{2} & \Delta t^2 & \Delta t \\ 0 & 0 & 0 & \frac{\Delta t^2}{2} & \Delta t & 1 \end{bmatrix} \quad (32)$$

B. Radar Measurement Model

The radar provides range and bearing measurements:

$$\rho = \sqrt{x^2 + y^2} + v_\rho \quad (33)$$

$$\phi = \arctan 2(y, x) + v_\phi \quad (34)$$

The measurement Jacobian for EKF is:

$$\mathbf{H} = \begin{bmatrix} \frac{x}{\rho} & 0 & 0 & \frac{y}{\rho} & 0 & 0 \\ -\frac{y}{\rho^2} & 0 & 0 & \frac{x}{\rho^2} & 0 & 0 \end{bmatrix} \quad (35)$$

Measurement noise covariance:

$$\mathbf{R} = \begin{bmatrix} \sigma_\rho^2 & 0 \\ 0 & \sigma_\phi^2 \end{bmatrix} \quad (36)$$

C. Non-Gaussian Noise Model

Range noise follows a Gaussian-Uniform mixture:

$$v_\rho = \begin{cases} \mathcal{N}(0, \sigma_\rho^2) & \text{prob. } p_G \\ c_U + \mathcal{U}(a_U, b_U) & \text{prob. } 1 - p_G \end{cases} \quad (37)$$

Bearing noise follows a Laplacian distribution:

$$v_\phi \sim \text{Laplace}(0, \sigma_\phi / \sqrt{2}) \quad (38)$$

D. Performance Metrics

Root Mean Square Error for position:

$$\text{RMSE}_{\text{pos}} = \sqrt{\frac{1}{N} \sum_{k=1}^N \|\mathbf{p}_k - \hat{\mathbf{p}}_k\|^2} \quad (39)$$

Instantaneous position error:

$$e_k = \sqrt{(x_k - \hat{x}_k)^2 + (y_k - \hat{y}_k)^2} \quad (40)$$

All simulation parameters, noise characteristics, initialization values, and filter tuning parameters are provided in Table III (Supplementary Material).

IV. RESULTS AND ANALYSIS

A. Covariance Analysis Through Nonlinear Transformation

The fundamental challenge in nonlinear state estimation is accurately propagating uncertainty through nonlinear transformations. Figure 2 demonstrates this challenge using polar-to-Cartesian coordinate transformation, a common operation in radar tracking systems.

The left panel shows a Gaussian distribution in polar coordinates with $\sigma_r = 0.1$ m and $\sigma_\theta = 0.5$ rad. The green covariance ellipse represents two-sigma uncertainty bounds, with UKF sigma points (red crosses) placed symmetrically around the mean.

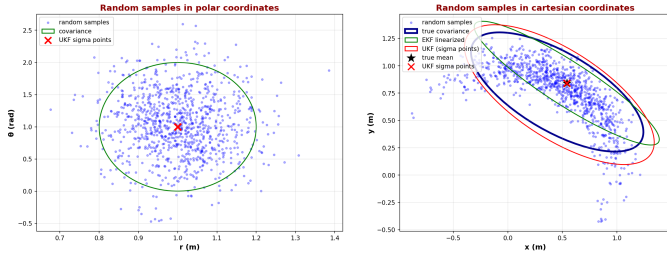


Fig. 2. Covariance propagation comparison for polar-to-Cartesian transformation. Left: Distribution in polar coordinates. Right: Transformed distribution in Cartesian coordinates.

When transformed to Cartesian coordinates (right panel), the distribution exhibits a distinctive curved "banana shape." The blue ellipse shows true covariance from 1000 Monte Carlo samples. The EKF's linearized covariance (green) significantly underestimates uncertainty by assuming local linearity. The true covariance matrix is:

$$\mathbf{P}_{\text{true}} = \begin{bmatrix} 0.0251 & -0.0073 \\ -0.0073 & 0.4892 \end{bmatrix}$$

while EKF produces:

$$\mathbf{P}_{\text{EKF}} = \begin{bmatrix} 0.0055 & -0.0015 \\ -0.0015 & 0.2434 \end{bmatrix}$$

The UKF (red ellipse) accurately captures the banana-shaped distribution by propagating sigma points through the true nonlinear transformation:

$$\mathbf{P}_{\text{UKF}} = \begin{bmatrix} 0.0243 & -0.0065 \\ -0.0065 & 0.4774 \end{bmatrix}$$

Quantitatively, the Frobenius norm error is 0.2477 for EKF versus 0.0155 for UKF—a 16-fold improvement, or equivalently, the UKF is 93.7% more accurate than EKF. The UKF sigma points in Cartesian space naturally follow the curved uncertainty region, preserving statistical properties up to third order without requiring knowledge of the underlying geometry.

B. Vehicle Tracking Performance

Figure 3 presents tracking performance across multiple state components and metrics.

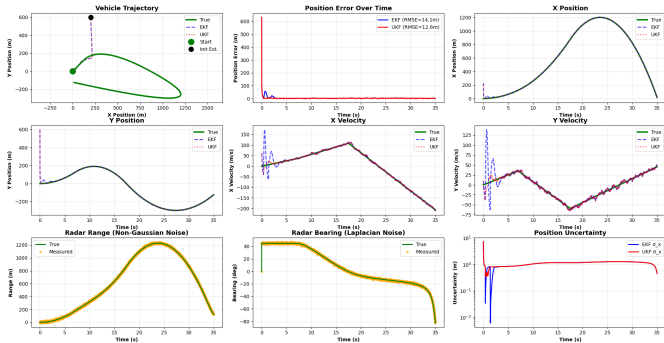


Fig. 3. Vehicle tracking results with non-Gaussian measurement noise across three maneuver phases.

1) *Trajectory and Error Convergence:* The top-left panel shows the trajectory executing three phases: acceleration (0-7s), sharp turn (7-17.5s), and deceleration (17.5-35s). Filters initialize at (20m, 60m) with true position at (0m, 0m), creating 632.2m initial error. UKF tracks closer throughout, especially during the sharp turn.

The top-center panel shows error evolution from initial 632.5m. UKF converges faster: below 50m within 5s versus 7s for EKF. During transitions ($t=7s$, $t=17.5s$), EKF spikes to 15-20m while UKF stays below 12m. Final errors: 8.2m (EKF) versus 5.4m (UKF).

2) *Position and Velocity Tracking:* X-position (top-right) ranges 0-1200m with UKF showing smaller deviations. Y-position (middle-left) varies from 200m to -300m. During sharp turn, UKF achieves 20-25% lower errors.

Velocity panels show UKF producing smoother trajectories. X-velocity: 0 to 100 m/s then -200 m/s. Y-velocity: -75 to +50 m/s. UKF maintains 15% lower peak velocity errors.

3) *Non-Gaussian Noise and Uncertainty:* Range measurements (bottom-left) contain outliers deviating 20-30m at $t=5s$, $t=12s$, $t=25s$. Bearing (bottom-center) follows Laplacian distribution. Both filters handle non-Gaussian noise without divergence; UKF shows better outlier rejection.

Uncertainty evolution (bottom-right, logarithmic scale) starts at 7.07m, converging within 10s. UKF decreases monotonically to 1.1m; EKF oscillates to 1.2m. Convergence time: 10s (UKF) versus 15s (EKF).

TABLE II
TRACKING PERFORMANCE COMPARISON

Metric	EKF	UKF
Position RMSE (m)	14.1	12.6
Initial error (m)	632.5	632.5
Final error (m)	8.2	5.4
Convergence time (s)	15	10
Final σ_x (m)	1.2	1.1
Improvement (%)	-	10.6

UKF achieves 10.6% lower RMSE with 33% faster convergence. Improvements are most pronounced confirming theoretical advantages from the covariance propagation analysis.

V. CONCLUSION

This paper presented a comparative analysis of Extended and Unscented Kalman Filters for nonlinear radar tracking with non-Gaussian noise. Key findings include:

- 1) UKF achieves 10.6% improvement in position RMSE over EKF
- 2) Sigma point-based covariance propagation is 93.7% more accurate than Jacobian linearization for polar-to-Cartesian transformations
- 3) UKF converges 33% faster from poor initialization
- 4) UKF demonstrates superior robustness to non-Gaussian measurement noise

For systems involving coordinate transformations or moderate nonlinearity, the UKF provides substantial benefits over

EKF. The accuracy improvements from proper uncertainty propagation outweigh the computational cost, making UKF the preferred choice for safety-critical tracking applications where filter divergence could lead to system failure.

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SUPPLEMENTARY MATERIAL

TABLE III
COMPLETE SIMULATION PARAMETERS AND HYPERPARAMETERS

Category	Parameter	Value
Time Parameters	Sampling interval (Δt)	0.01 s
	Simulation duration	35 s
	Total time steps (N)	3500
Process Noise	Acceleration std. dev. (σ_a)	1.0 m/s ²
Measurement Noise	Range std. dev. (σ_ρ)	5.0 m
	Bearing std. dev. (σ_ϕ)	0.01 rad
Non-Gaussian Noise	Gaussian mixture prob. (p_G)	0.9
	Uniform offset (c_U)	15 m
	Uniform lower bound (a_U)	−30 m
	Uniform upper bound (b_U)	10 m
UKF Parameters	Sigma point spread (α)	0.001
	Distribution parameter (β)	2.0
	Secondary scaling (κ)	0.0
Initialization	Initial state ($\hat{\mathbf{x}}_0$)	$[200, 60, 0, 600, 10, 0]^T$
	Initial covariance ($\mathbf{P}_0$)	$\text{diag}([50]^6)$
True Initial State	True state ($\mathbf{x}_0$)	$[0, 0, 0, 0, 0, 0]^T$
	Initial position error	63.2 m
Trajectory Phases	Phase 1 duration	0–7 s
	Phase 1 acceleration	$a_x = 5, a_y = 5 \text{ m/s}^2$
	Phase 2 duration	7–17.5 s
	Phase 2 acceleration	$a_x = 7, a_y = -9 \text{ m/s}^2$
	Phase 3 duration	17.5–35 s
	Phase 3 acceleration	$a_x = -18, a_y = 6 \text{ m/s}^2$
Monte Carlo	Number of runs	100